covtl-sandhi v0.1.0

A Didactic Guide to Inter-Word Sandhi
in the COVTL Pipeline Languages (fr, en, it, es, pt, de)

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Abstract

This guide documents `covtl-sandhi` v0.1.0, the standalone package of *inter-word sandhi* rules for the six languages of the `covtl-pipeline` articulatory speech-synthesis pipeline. It is written for a pipeline user who knows phonetics but not this code base. We explain what inter-word sandhi is and why a strictly word-by-word grapheme-to-phoneme (g2p) conversion can never produce it, how the package is wired into the pipeline through the two-branch architecture (hook + rules package), how the three historical modes (`g2p`, `post`, `pre`) differ in fidelity, and — chapter by chapter — every rule that each of the six language modules actually applies, with step-by-step derivations from orthography to engine SAMPA. Coverage levels, editable data tables, known limits, installation and verification complete the picture. Every rule and every example cited here is taken from the package sources (README, CHANGELOG, module docstrings, tests) and from the `README_LIAISON.md` of the historical five-language patch; nothing is invented. SAMPA transcriptions are set in monospace.

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Chapter 1

Introduction: What Is Inter-Word Sandhi?

1.1 The phenomenon

Sandhi is the family of phonological modifications that happen at boundaries between linguistic units. *Inter-word sandhi* (French *liaison*, cross-word sandhi) is the part of it that happens *at word boundaries inside a phrase*. Six examples, one per language of the package:

language	phenomenon	example	isolated → connected
fr	latent consonant (liaison)	"faut il"	`fo.i` → `fo.til`
en	a→an + onset linking / linking-r	"far away"	`faR.@.we` → `fa.R@.we`
it	raddoppiamento sintattico + euphonic *d*	"a casa"	`a.ka.sa` → `a.kka.sa`
es	enlace + y→e / o→u	"los amigos"	`los.a.mi.gos` → `lo.sa.mi.gos`
pt	encadeamento + EP voicing	"os amigos"	`oS.@.mi.guS` → `o.Z@.mi.guS`
de	Glottalversatz (anti-liaison)	"ein Apfel"	`ajn.a.pf@l` → `ajn a.pf@l`

In every case the modification is triggered by *the pair* (the left word **W1** followed by the right word **W2**): a consonant that is written but mute at the end of W1 surfaces as the onset of W2 (French), an article takes *n* before a vowel (English), an initial consonant geminates after a trigger word (Italian), a final consonant migrates into the onset of the next syllable (Spanish, Portuguese), or — the deliberate mirror image — a glottal boundary *separates* the two words instead of linking them (German).

1.2 Why a word-by-word g2p cannot produce it

The pipeline — like every text-to-SAMPA front end built as a per-word transducer — converts *one word at a time*. This is by design: a clean per-word g2p (lexicon → exceptions → rules) is testable, cacheable and maintainable. But the information needed for inter-word sandhi simply *does not exist inside a single word*:

- the latent consonant of French *faut* is mute in isolation (`fo`); nothing in the isolated word tells you that a `t` must appear when *il* follows;
- the Spanish final `s` of `los` is a coda in isolation; only the vowel-initial `a.mi.gos` on its right turns it into an onset;
- the Italian trigger status of *città* (written final accent) is a property of W1, but the gemination lands on W2.

The guide's opening example says it all: a perfect per-word g2p gives `fo.i` for "faut il"; the correct connected form is `fo.til`. Whatever is missing lies *between* the two per-word tokens — exactly where a word-by-word transducer never looks. This is why the pipeline needs a separate, pair-aware layer: the `covtl-sandhi` package documented here.

1.3 Where covtl-sandhi sits in covtl-pipeline

`covtl-sandhi` lives strictly in the **text/g2p layer** of the pipeline, next to the expressive-prosody patch. It rewrites text or SAMPA *before* the synthesis engine is called. Three architectural invariants hold everywhere (module docstrings, README, CHANGELOG):

1. **The synthesis engine is never modified.** The engine modules (`syltraj`, `polar`, `gesture`, `build_phrase_tract`, ...) receive an ordinary SAMPA string. It was verified on a pipeline clone that linked forms such as `mO~.na.mi` pass `syltraj` without the `V~.V` nasal collapse: the linking consonant interposes itself between the two vowels.

2. **Every language is respected.** Each rule table applies only to word pairs of its own language; any other profile is a no-op with a warning. The English profile is covered by its own module (new in v0.1.0), not ignored.

3. **Compatibility with the expressive-prosody patch.** The sandhi key (inserted or displaced) is always a *consonant*, never a syllabic nucleus, and nothing is ever inserted across a pause (`%`, `|`, comma). Section 3.5 explains why this matters for the F0 plans.

1.4 Notation and how to read this guide

Engine SAMPA is set in monospace: `fo.til`. Within a phrase string the engine uses the following separators (mirrored from the pipeline g2p conventions, cf. the module docstrings):

symbol	meaning
`.`	syllable link (inside a word, and between linked words of a phrase)
`(space)`	word boundary — a short pause / re-articulation
`|` (with spaces)	sentence boundary
`%`	breath pause inserted by the expressive-prosody plan

Chapters 2–3 cover architecture and method; Chapters 4–9 are one chapter per language, each opening with a spoken example and then listing *all* trigger classes, exceptions and derivations the module implements; Chapters 10–13 cover coverage levels, editable tables, limits, and installation. Sources are cited inline as file names; the phonological references of each module are collected in the bibliography.

Chapter 2

Architecture: Branch A (Hook) + Branch B (This Package)

2.1 Overview

The distribution splits cleanly into two branches that can be installed independently:

- **Branch A — the hook**: a light git patch `patch/covtl-pipeline-sandhi-hook.patch` applied to the *pipeline* repository. It adds a stub package `vtl_synth/utils/sandhi/` (with `SANDHI_FULL = False`: every entry point is the identity) and the command-line options `--sandhi` and `--sandhi-mode`.
- **Branch B — the rules**: the `covtl-sandhi` package itself (this distribution): six language modules (`fr.py`, `en.py`, `it.py`, `es.py`, `pt.py`, `de.py`), the shared `base.py`, the editable views `data/*.txt`, and the standalone test suite.

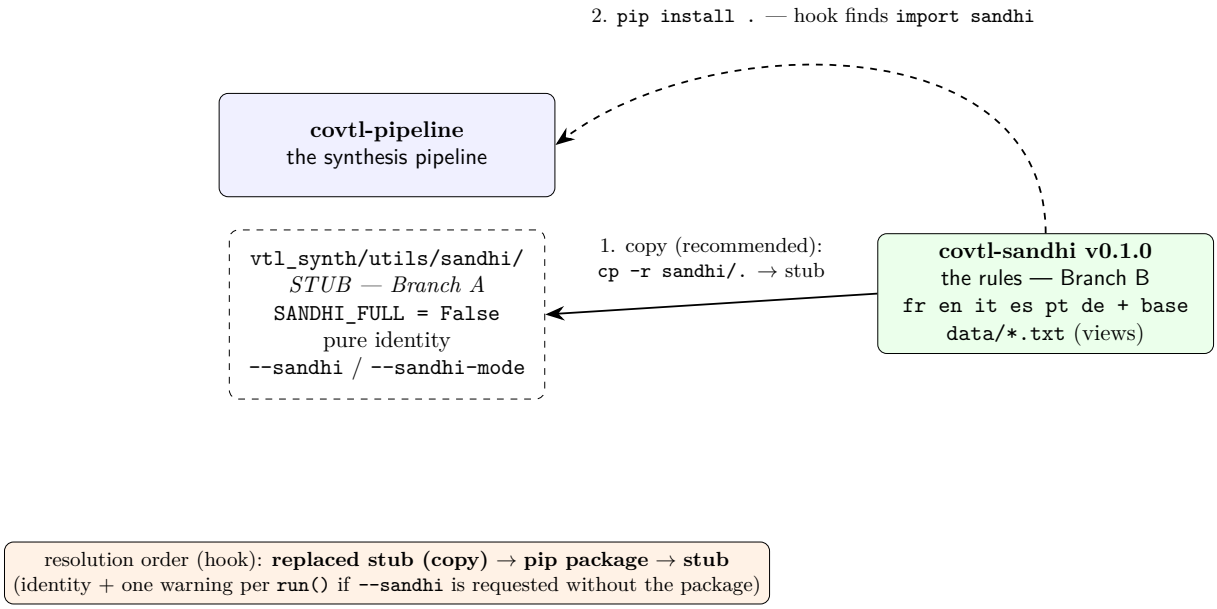


Figure 2.1: Branch A / Branch B architecture and the hook's resolution chain.

2.2 Branch A: the hook, and the bit-identical guarantee

The hook patch is deliberately boring: the stub it installs contains the same function signatures as the full package, but every function is the identity (`SANDHI_FULL = False`), and the new CLI options default to `--sandhi off`. The consequence, verified on a clean clone of the pipeline (`main` @ `41f6571`, *INSTALL.txt*):

- the patch applies cleanly; the repository test suite goes from **155 passed** (virgin clone) to **165 passed** with the hook — exactly the +10 hook tests, **0 regressions**, the 3 pre-existing failures unchanged;
- with the hook alone the pipeline outputs are **bit-identical** to the unpatched repository: the hook is invisible by default.

This is what makes the hook safe to merge and ship independently of the rules: nothing changes until Branch B is installed *and* the user asks for `--sandhi`.

2.3 Branch B: activating the rules

Two equivalent activations (pick one):

1. **Direct copy (recommended to stay self-contained).** The stub folder *already exists* after the hook, so you copy the **content** of the package's `sandhi/` directory — note the trailing `/.` in

   ```
   cp -r sandhi/.  <pipeline>/vtl_synth/utils/sandhi/
   ```

   Without the `/.`, `cp -r sandhi/ dst/` would nest the folder one level too deep (`dst/sandhi/...`) and the hook would keep finding its stub. (PowerShell equivalent: `Copy-Item -Path sandhi\* -Destination dst -Recurse -Force`.) The package overwrites the stub; the hook now reads `SANDHI_FULL = True`.

2. **pip.** `pip install .` installs the package into site-packages; if the stub has not been replaced, the hook detects the importable `sandhi` package instead.

The full resolution chain is therefore: replaced stub (copy) → pip package → stub (identity, plus one warning per `run()` if a sandhi level was requested without any package installed).

Chapter 3

The Method Inside the Pipeline

3.1 The per-word oracle

Every module exposes an *oracle*: a function `oracle(word, g2p) -> token` that returns the SAMPA token of one word. The oracle is the package's only window onto the pipeline's g2p: the active g2p is called word by word through its public interface `(text, warnings) -> str` — there is no import from any `g2p_*` module (independence requirement of the framework).

- For **it, es, pt, de, en** the per-word g2p is backed by a lexicon (Wikipron for it/es/pt/de, CMUdict for en) and was measured exact on the clone: the oracle is the *identity*. It exists for architectural symmetry.
- For **fr** the base g2p damages the closed class (*il* → `i`, *est* → `e`, *petit* → `ti`, *plus* → `py`…). The French oracle applies a small table of corrections, `_PRON_OVERRIDES` (`sandhi/fr.py`), *only to words that belong to a liaison pair* — the rest of the phrase keeps the base tokens, so `off` stays bit-identical:

word	override	base g2p	comment
il / ils	`il`	`i`	final `l` recovered
elle	`El`	`a`	
est	`E`	`e`	open vowel
cet	`sE`	`st`	
ces	`se`	`se.@.Es`	
petit	`p@.ti`	`ti`	
plus	`ply`	`py`	
second	`s@gO~`	`zgO~`	
huit	adds final `t`	loses it	initial glide kept
dernier	`dER.nje`	`dERn.je`	
ancien	`a~sjE`	`a~sjE~`	denasalised
certain	`sER.tE`	`sERtE~`	denasalised

3.2 Pair planning: from text to pairs

All modules share the same text-splitting skeleton (a fresh mirror of each language g2p's conventions; see the regex tables in the modules): sentences are split on `[.!?;:]+`, comma-parts on `[,-]` (a comma is a boundary: *a pair never links across it*), words by a per-language letter class (French keeps the apostrophe — *c'est*; Italian, Spanish, Portuguese and German do not, mirroring their g2p tokenisers).

`plan_pairs(text, g2p, level)` then walks every adjacent pair (`W1`, `W2`) of every comma-part, asks the module's decision function *"does this pair carry a sandhi, and at which coverage level?"* and returns the plan: for each pair index, either `None` or a record (kind, key to insert or displace, both tokens). Pairs may *chain*: in Italian "così come va" both così→come and come→va fire; in Spanish "es un amigo" the word *un* both receives `s` and loses its `n`.

3.3 SAMPA fusion in slots

The fusion step (`apply_liaison_post` / `apply_sandhi_post` in each module) first *aligns* the plan against a produced SAMPA string: sequentially, each word's token is found as a slot delimited by `.` or space (`_find_slot`); internal dots of a token (`a.mi.gos`) belong to the token and never create false slots; the alignment must consume the sentence exactly (nothing left except `%` markers). Then, right-to-left so offsets stay valid, each planned pair is fused:

- **insertion** (fr latent, it geminate/d, en *an*): the key is carried by the *right slot*, in onset — `['fo','i']` → `['fo','ti']`;
- **displacement** (es enlace, pt encadeamento, en onset linking): the left slot is amputated of its final consonant, which becomes the onset of the right slot — `['los','a.mi.gos']` → `['lo','sa.mi.gos']`; chained groups are reconstructed segment by segment;
- **boundary** (de): the right slot receives a *leading space* — `['ajn','a.pf@l]` → `['ajn',' a.pf@l']`, and the join `join_words_keys` does not double the separator.

If the gap between the two slots of a pair is anything but `.` or a space — i.e. a `%` breath pause, a `|` sentence boundary, or a comma — the pair is skipped. On any alignment failure the whole sentence is returned unchanged (fail-safe, never a wrong form).

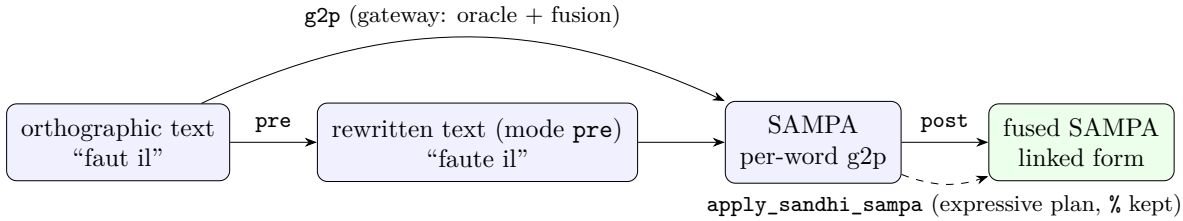


Figure 3.1: The three modes and the expressive path. `g2p` is the recommended route; `pre` rewrites the text and feeds the plain g2p; `post` edits an already produced SAMPA.

3.4 The three modes and their fidelity trade-offs

The option `--sandhi-mode` keeps the three historical implementations of the five-language patch as a user choice (`README_LIAISON.md`, module docstrings):

mode	where it acts	fr “faut il”	strengths / limits (measured)
`pre`	before the g2p: orthographic rewrite of the text (“faut” → “faute”); g2p untouched	`fot.i`	maximally decoupled; latent lands in *coda* (not onset), small vowel symbol shifts (`o~`/`O~`), partial coverage (no correct respelling for *est*, *petit*, *plus*, *vieux*, *ces*, *plein*, *jamais*, *beaucoup*); no-op for de/en (no spelling exists).
`g2p` **(default, recommended)**	in the g2p layer: sandhi-aware gateway — per-word oracle + pair fusion with corrected tokens	`fo.til`	latent in *onset*, closed-class corrections, chains with `--expressive`; the only insertion point that sees both the pair and the pronunciation.
`post`	after the g2p: pure SAMPA post-processor (insertion only, base g2p totally untouched)	`fo.ti`	most decoupled after the fact; inherits the base g2p per-word defects (here *il* → `i`: the `l` is missing — inserting cannot repair a token).

The method search that concluded with this design is quoted in `README_LIAISON.md`: the word boundary is visible *neither* upstream — spelling does not reliably encode the phenomenon (and German encodes nothing at all) — *nor* downstream — the produced SAMPA has lost the boundary, and the g2p's per-word defects cannot be repaired by insertion. The g2p layer is the only place where both are available. For it/es/pt/de/en the oracle is the identity, so `g2p` and `post` produce identical outputs there; the difference is architectural. In `pre` mode the gateway returns the g2p output *over the rewritten text*; `sandhi_rewrite_text` returns the rewritten orthography alone, so the expressive planner can be fed the same respelled text as the monotone path.

3.5 The expressive path and the key invariant

The expressive-prosody patch plans F0 over the phrase *before* the final synthesis, and its plan indexes syllable *nuclei*. The expressive entry point of the package is therefore

```
apply_sandhi_sampa(lang, text, sampa, g2p, level, mode)
```

which merges the planned pairs into an *already produced*, breath-marked SAMPA (`%` markers present). With `mode='g2p'` the planner saw the oracle tokens, so the pair slots are rewritten with the corrected tokens; with `mode='post'` the planner saw the base g2p and only the insertion is done. Two properties are guaranteed and tested:

1. **Never across a pause.** A pair whose slots are separated by `%`, `|` or a comma is left alone — a breathing boundary is phonologically stronger than any sandhi (test `test_expressive_jamais_a_travers_pause`: with `'mO~.%.a.mi'` the `%` survives; without it the fusion gives `'mO~.na.mi'`).

2. **The key is a consonant, never a nucleus.** Every key the package inserts or displaces is a consonant (or, in German, a boundary that separates existing segments). A consonant

inserted into an onset adds no syllable and removes none, so the *number and order of nuclei* — the anchors of the F0 plans — stay aligned. This is why vowel-fusion phenomena (Spanish *sinalefa*, English *smoothing*, Portuguese atone elision) are deliberately out of scope: they would delete a nucleus and desynchronise the F0 tiers. The framework's phrase for it: *a sandhi key is never a nucleus.*

Chapter 4

French — the Latent Consonant (Liaison)

"faut il" → `fo.til`

Spoken in isolation, *faut* ends on the vowel `fo`; the written final `t` is mute. Before the vowel-initial pronoun *il*, that latent consonant surfaces — not as a coda `fot.i`, but as the *onset of the next syllable*: `fo.til`. That placement (TTS usage; Delattre 1951, Encrevévé 1988, Tranel 2000) is the module's signature.

4.1 The rule the module implements

The module (`sandhi/fr.py`) decides a liaison for the pair (*W1*, *W2*) when *all* of the following hold:

1. *W1* belongs to a trigger class (below) — or, at level `careful` only, is a plausible plural noun/adjective;
2. the token of *W2* starts with a vowel (SAMPA test — a *mute h* like *heureux* `2.R2` counts as vowel-initial);
3. *W2* is not an *h aspiré* word (closed list) and not a proper noun (capitalised, not part-initial);
4. *W1* is not *et* (the classical blocker);
5. the token of *W1* does not already end with the latent phone (*sept heures* `sE.t9R`: natural link, no double consonance);
6. no sentence or comma boundary between the two words.

The latent phone is read from the *last written letter* of *W1*, with standard devoicing and the *neuf* exception:

written letter	t	d	s	x	z	n	r	p	f	g
latent phone	`t`	`t`	`z`	`z`	`z`	`n`	`R`	`p`	`f`	`g`

Written `d` gives voiceless `t` (*grand ami* → `gRa~.ta.mi`, never `[d]`). *neuf* is special: latent `v` before *a-* (*neuf ans* `n2.va~`), latent `f` elsewhere (*neuf heures* `n2.f9R`).

4.2 Trigger classes and levels

- **minimal — obligatory liaisons** (categorical): determiners, possessives, demonstratives (*les, des, ces, cet, cette, mes, tes, ses, nos, vos, mon, ton, son, un*); pronouns (*il, ils, elle, elles, on, en, nous, vous*); copulas and auxiliaries *être*/*avoir*, 3rd person (*est, sont, ont, avait, avaient, était, étaient, étais, étaient, es*).
- **standard — the everyday class** (facultative but normal) adds: modals and full verbs of the closed list (*faut, doit, peut, veut, font, vont, dit, fait, vais, c'est, n'est, est-ce*); adverbs and prepositions (*très, trop, plus, moins, bien, souvent, pendant, quand, sans, dans, chez, jamais, comment, pas, tout, tous, mais, rien, combien, avant, assez, vraiment, dont*); preposed adjectives (*petit, grand, gros, bon, mauvais, premier, dernier, second, ancien, certain, vingt, cent, plein, long, vieux, ancienne, grande*); numerals (*deux, trois, six, dix, neuf, beaucoup*).
- **careful — style soutenu**: liaison after a plural noun or adjective. Because French spelling does not mark the noun/verb boundary, this is a *closed heuristic*: *W1* of length ≥ 3 ending in `-s` or `-x`, excluding detectable verbal endings (`-es, -ent, -ez, -ons`). *partent encore* never links, even at `careful` (test `test_fr_careful_exclut_formes_verbales`); real invariables such as *puis*, *toujours* remain admissible.

The *h aspiré* blocking list is an editable view, `data/fr_h_aspire.txt`: *héros, haut, hache, haie, haine, hall, halle, harpe, hasard, hélas, hectare, homard, huche, hoyau, hamburger*. Note the contrast inside the module: *haricot* is *not* in the list — modern usage treats its *h* as mute, so *les haricots* links normally.

4.3 Worked derivations

Each derivation shows: pair detected → level of the firing rule → latent phone → slot fusion (g2p mode; oracle tokens).

"les amis" → `le.za.mi` (minimal).
Pair (*les*, *amis*): *les* determiner → minimal; token `a.mi` vowel-initial; last letter `s` → latent `z`; `le` does not end in `z`. Fusion `le` + `.z` + `a.mi` → `le.za.mi`.

"mon ami" → `mO~.na.mi` (minimal).
Possessive *mon*; latent `n` from the written `n`; fusion `mO~` + `.n` + `a.mi`. The nasal vowel is denasalised exactly as the phonology requires. (Mode `pre` respells *mon* → *monne*: the latent lands in coda, `mo~n.a.mi`, with the documented `o~`/`O~` shift.)

"vous avez" → `vu.za.ve` (minimal).
Pronoun *vous*; written `s` → `z`; `vu` + `.z` + `a.ve`.

"faut il" → `fo.til` (standard; oracle at work).
faut is a modal → standard class (at `minimal` the pair does not fire: test `test_fr_minimal_exclut_courante`). Oracle tokens: *faut* → `fo`, *il* → `il` (correction from the base `i`!). Latent `t`; fusion `fo` + `.t` + `il` → `fo.til`. In `post` mode the base token `i` survives: `fo.ti` — insertion cannot repair a token.

"grand arbre" → `gRa~.taRbR` (standard).
Preposed adjective *grand*; written `d` devoices to `t`; `gRa~` + `.t` + `aRbR`.

"très intelligent" → `tRE.zE~te.liZa~` (standard).
Adverb *très* → `tRE`; latent `s`→`z`; *intelligent* → `E~te.liZa~`; result `tRE.zE~te.liZa~` — the syllables merge around the latent consonant.

“très heureux” → `tRE.z2.R2` (standard).
Same rule; *heureux* has a *mute h* (`2.R2`, vowel-initial) → the liaison crosses it.

“neuf heures” → `n2.f9R` / “neuf ans” → `n2.va~` (standard).
The *neuf* exception: `f` before a non-*a* vowel, `v` before *a*-.

“comment allez-vous” → `kOma~.ta.le.vu` (standard).
The historical repository case (the `syltraj` `V~.V` collapse): the latent `t` of *comment* repairs the boundary.

“les haricots” → `le.za.Ri.ko` (minimal).
haricot is not on the *h aspiré* list (mute *h* in modern usage) → normal liaison.

“les héros” → `le.e.Ro` — *blocked*.
héros is on the list: no liaison at any level (test `test_fr_h_aspire_blocker`).

“les enfants” → `le.za~.fa~` (minimal).
Latent `z` + nasal vowel: the onset consonant cuts the `a~.fa~` collapse.

“est-il” → `E.til` (minimal, g2p mode).
Closed-class corrections on *both* sides: *est* → `E`, *il* → `il`; latent `t`. Mode `pre` has no respelling for *est* (documented): the pair stays unlinked, with the warning `no fallback spelling for 'est'`.

“quand il” → `ka~.til` (standard).
Nasal vowel + latent `t`; verified synthesizable by `syltraj` (no collapse).

“enfants adorables” → careful only: `A~.fa~.za.dO.rabl`.
At `standard`: no liaison (*enfants* is not a trigger). At `careful`: plural heuristic `-s`, not a verbal ending → latent `z` fires (test `test_fr_careful_pluriel`).

“les amis, cherchent encore”.
Liaison inside the first comma-part only (`le.za.mi`); *never* across the comma — and *cherchent* (`-ent`, verbal) is not a trigger even at `careful`: `le.za.mi.SErS.a~.kOR` (test `test_fr_pas_a_travers_virgule`).

“vous avez un très grand arbre” + `--expressive`.
Two liaisons (*vous avez*, *grand arbre*) composed with the expressive plan; no insertion ever crosses the `%` breathing pause of the plan.

4.4 Mode `pre`: the measured respelling table

`pre` replaces the *carrier* word by a fallback spelling that the *unmodified* g2p converts to [pronunciation + latent]. Every entry was measured on the clone and is re-verified at runtime (a drifted entry is ignored with a warning): *faut*→*faute* `fot`, *grand*→*grante* `gRa~t`, *mon*→*monne* `mo~n` (vowel shift accepted), *vous*→*vouze* `vuz`, *les*→*léze* `lez`, *deux*→*deuze* `d2z`, *très*→*trèze* `tREz`, *quand*→*quante* `ka~t`, *bon*→*bonne* `bOn`, *neuf* → *neuve/neufe* depending on the following vowel, and so on (the full table is `_PRE_RESPELL` in `sandhi/fr.py`). Known gaps (no correct fallback): *est, petit, en, plus, vieux, ces, plein, jamais, beaucoup* — those pairs warn and stay unlinked in `pre`.

Chapter 5

English — a → an and Onset Linking (Linking-r)

"a apple" → `@.na.p@l` "far away" → `fa.R@.we`

The English module (new in v0.1.0, `sandhi/en.py`) mirrors two Romance phenomena of the package and measures everything on the pipeline's CMUdict gateway.

5.1 a → an — a categorical, *phonetic* rule

When *W1* is the bare indefinite article *a* and the *token* of *W2* begins with a vowel, the article takes `n`, carried by the right slot in onset — the exact mirror of the French latent consonant: `('@','a.p@l')` → `('@','na.p@l')` → `@.na.p@l`.

The test is phonetic, not orthographic, because CMUdict has already resolved the spelling: *hour* → `a.w@R` (vowel → *an hour*), *one* → `w@n` (consonant → *a one way*, no `n`), *university* → `ju...` (initial glide → *a*). All three behave correctly *by construction* (tests `test_en_an_test_phonetique`). Level: **minimal** (categorical, exceptionless).

5.2 Onset linking, and why linking-r is not a separate rule

The final consonant of *W1* — a coda when the word stands alone — becomes the onset of the first syllable of a vowel-initial *W2*. The consonant must be *simple and movable*: the last syllable must keep a nucleus after amputation (`'s'` alone is refused), glides `j`/`w` are never displaced, and affricates `tS`/`dZ` move as one key (`_split_final_consonant`).

Because the CMUdict gateway follows British-lexicon conventions, the non-pre-vocalic `r` is *already* in the token (*far* → `faR`): linking-r is simply onset linking applied to a token ending in `R` — the module has no separate rule for it. Dynamic stress is untouched: the displaced onset does not move the word's written stress (*far away* keeps its stress on *way*).

Level: **standard** (productive). **careful = standard**, stated honestly: at the token level the CMUdict gateway offers no reliable register-dependent rule.

5.3 Worked derivations

"a apple" → `@.na.p@l` (minimal).
a → `@`; *apple* → `a.p@l` vowel-initial; insert `n` in onset: `na.p@l`.

"far away" → `fa.R@.we` (standard).
far → `faR` splits into (`fa`, `R`); *away* → `@.we` vowel-initial; the displaced `R` becomes the onset: `fa.R@.we`. (At `minimal`: unchanged `faR.@.we`.)

“turn off” → `t@R.nOf` (standard).
`t@Rn` → (`t@R`, `n`); the `n` links.

“car engine” → `ka.REn.dZ@n` (standard).
`kaR` → (`ka`, `R`); the `R` attaches to `En.dZ@n` → `REn.dZ@n`.

“her own car” → `h@.Ron.kaR`.
Docstring example: `h@R` → (`h@`, `R`); *own* → `on`.

“the end” → `D@.End` — *no link*.
the → `D@` ends in a vowel: there is no consonant to displace (test `test_en_pas_de_link_apres_voyelle`). Enlace without a consonant does not exist in the module.

5.4 Mode `pre` in English

`a`→`an` has a real spelling (the base g2p already knows *an* `an`, verified by round-trip), so `pre` rewrites it: “a apple” → “an apple”. Onset linking has *no spelling* (the *r* of *far* is written; the displacement is phonetic), so each linking pair warns — `no onset linking spelling - use g2p or post` — and the text is left intact. *Smoothing* (“go away” → [g@.we]) is out of scope: vowel fusion would delete a nucleus (Section 3.5).

Chapter 6

Italian — Raddoppiamento Sintattico + Euphonic d

"a casa" → `a.kka.sa` "e io" → `e.di.o`

Italian "liaison" is not a latent consonant. It is two sandhi, both essentially invisible in spelling (`sandhi/it.py`, after Serianni 1989 and Bertinetto & Loporcaro 2005).

6.1 Raddoppiamento sintattico (RS)

After a *trigger* word, the initial consonant of the next word geminates, in onset — exactly the `cluster_onset` convention of the pipeline's own Italian g2p (`bE.lla`), so the engine needs nothing new. Triggers are:

1. **every oxytone** — Italian spelling *always* marks a stressed final vowel (*città, però, virtù, caffè, sarà, andò, è, più, può*...): detection is a purely graphic, deterministic rule (final letter in `à è é ì ò ú í ú`);
2. **a closed class of function words without written accent** (Serianni's list, editable in `data/it_rs_triggers.txt`): *a, da, e, o, ma, se, che, come, dove, qualche, fra, tra, su, fu, qui, qua, sta, sto, va, fa, do.*

Non-triggers (the norm): *per, con, in, di, non*, articles and atonic pronouns. Guards: the gemination is cancelled when *W2* is vowel-initial (*però il*: the RS vanishes — *però* is not euphonic either); it is deferred before glides `j`/`w` (variable norm); it never applies when the left token ends in a consonant (no triple consonant); and there is *no proper-noun blocking* — RS is a phonetic consequence, not a grammatical rule (*a Roma* → `a.RRo.ma`, docstring) — a documented difference from the French patch. Affricates geminate by lengthening the *occlusion*: `tS` → `t.tS`, `dZ` → `d.dZ`, the mirror of the g2p convention *zz* → `t t s`.

6.2 Euphonic d

Before a vowel-initial word (orthographic test), the bare conjunctions take *d*: *e* → *ed*, *a* → *ad*; *o* only before *o-* (*od oggi*, classical usage). When the *d* is already written (*ed anche*), it is not inserted — it *geminate*s like any initial after the trigger *e*: both "e anche" and "ed anche" end in the same sandhi `[ed:]`. *è* is not euphonic (*è ora*: never **èd ora*).

6.3 Worked derivations

"a casa" → `a.kka.sa` (minimal).
a monosyllabic trigger; *casa* → `ka.sa`, initial `k` geminate key; `a` ends in a vowel; fusion `a` + `.k` + `ka.sa`. (`pre` doubles the letter: "a ccasa", which the rule table reads `kka.za` — the voiced Tuscan intervocalic `s`, an accepted tolerance.)

"città bella" → `tSi.tta.bbE.lla` (standard).
Oxytone trigger (written à); at `minimal` (monosyllabic triggers only) the pair does not fire: `tSi.tta.bE.lla`.

"come va" → `ko.me.vva` (standard).
come is polysyllabic → standard (at `minimal`: `ko.me.va`).

"e io" → `e.di.o` (minimal).
W2 vowel-initial → euphonic-d branch; bare *e* → `d` in onset: `e` + `.d` + `i.o`. (`pre`: real spelling "ed io".)

"ed anche" → `ed.da.nke`.
Written *ed*: the `d` geminates after the trigger *e* — same result as "e anche" (bare *e* + inserted `d`). `pre` has no spelling for this gemination (documented).

"a zero" → `a.ddzE.Ro`.
Initial affricate `dz`: the geminate lengthens the occlusion (`d` + `dZ`). `pre` cannot write it ("azzero" would read back `t t s`) → pair deferred, warning.

"a gente" → `a.ddZE.nte`.
Same occlusion-lengthening for `dZ` (demo page).

"così come va" → `ko.si.kko.me.vva`.
Chained pairs: così→come (`k`) and come→va (`v`); both fusions compose.

"andò via" → `a.ndO.vvi.a` (g2p/post).
A documented `pre` limit: the Wikipron lexicon transcribes *via* with hiatus (`vi.a`) while the rule table reads a diphthong — the round-trip fails, the pair stays ungeminated in `pre`.

Levels: **minimal** = euphonic *d* + monosyllabic RS triggers; **standard** adds *come*, *dove*, *qualche* and all written oxytones; **careful = standard** — no Italian rule of this set is register-dependent (documented honestly).

Chapter 7

Spanish — Enlace + Conjunction Alternation (y → e, o → u)

"los amigos" → `lo.sa.mi.gos` "y Ignacio" → `e.i.gna.Tjo`

Two phenomena (`sandhi/es.py`, after RAE 2009 §28.4/§26.6 and Navarro Tomás 1918).

7.1 Enlace — displacement of the final consonant

The final consonant of *W1* (simple, movable, not a glide; the last syllable must keep its nucleus) migrates to the onset of a vowel-initial *W2*. The base g2p already joins words with `.` but leaves the consonant in coda (`los.a.mi.gos`); the enlace moves it one syllable to the right. The *stress follows the linked syllable*: *un águila* → `u.na.gi.la` — the written accent of the `pre` respelling (*u náguila*) codes exactly that displacement. Chained pairs reconstruct *segment by segment* (each word emits [consonant received] + [token amputated/alternated/intact]), so *es un amigo* composes both links. Level: **standard**.

7.2 Conjunction alternation

The bare conjunctions alternate to avoid hiatus (RAE normative spelling): *y* → *e* before an *i-*/*hi-* initial, *o* → *u* before *o-*/*ho-*. The alternated form is a real spelling the g2p already knows; the module applies it when the source text carries the bare form. Level: **minimal**.

7.3 Worked derivations

"los amigos" → `lo.sa.mi.gos` (standard).
`los` splits into (`lo`, `s`); `a.mi.gos` vowel-initial; left slot amputated, right slot receives the onset: `lo` + `.s+a.mi.gos`. (At `minimal`: unchanged `los.a.mi.gos`.) `pre`: "lo samigos", verified by round-trip.

"un águila" → `u.na.gi.la` (standard).
`un` → (`u`, `n`); stress follows the new linked syllable `na`. `pre`: "u náguila".

"es un amigo" → `e.su.na.mi.go` — chained.
Group reconstruction: *es* emits `e` (its `s` moves), *un* emits `su` (receives `s`, loses its `n`), *amigo* emits `na.mi.go`. `pre` composes the same edits: "e su namigo".

"y Ignacio" → `e.i.gna.Tjo` (minimal).
Alternation: *Ignacio* starts with *I-* → *y* replaced by the token of *e*. No further rule (*Ignacio* is consonant-initial in SAMPA, `i.gna.Tjo` begins with the glide `i` — no enlace key).

“o otro” → `u.o.tRo` (minimal).
o → *u* before *o*-.

“por amor” → `po.Ra.mor` (standard).
`poR` → (`po`, `R`). `pre` fails here: the palatal `R` has no spelling → chain unlinked with a warning (documented).

“son hermanos” → `so.ne.Rma.nos` (standard).
The mute *h* is already resolved by the g2p (*hermanos* → `eR.ma.nos`): the enlace crosses the silent letter without any special rule.

“de agua” → `d@.a.ga` — *no pair*.
de ends in a vowel: nothing to displace. *Sinalefa* (vowel fusion) is out of scope — it would remove a nucleus from the F0 plans.

Expressive composition.
Los amigos y las alas with `--expressive`: three enlaces plus the alternation, F0 tiers aligned (the displaced consonant is never a nucleus).

careful = standard (no register-dependent rule at token level; documented).

Chapter 8

Portuguese — Encadeamento + European-Portuguese Voicing

"os amigos" → `o.Z@.mi.guS`

Structurally the Spanish mechanism — a final consonant migrates into the onset — with one Portuguese-specific variation (`sandhi/pt.py`, after Mateus & d'Andrade 2000 and Câmara 1970).

8.1 The rules

- **Encadeamento**: the final *simple* consonant of *W1* (`s`, `r`, `l`, `d`, `z`…) becomes the onset of a vowel-initial *W2*. Final nasals are *vocalised* in Portuguese (`u~`, `ko~`): they hold no movable consonant, so such pairs are a deliberate no-op — the careful form (*um elefante* stays `9~ @.l@fa~t`).
- **Linking voicing** (EP/BP): the final `/s/` voices when it links — table `{S: Z, s: z}`. The pipeline's Portuguese g2p uses the Lisbon palatal convention for the pausal form (*os* → `oS`, *amigos* → `a.mi.guS`, both with a pausal palatal `S`), so the linked form of *os amigos* is `o.Z@.mi.guS` — the `S` of the article surfaces as its voiced counterpart `Z` in the onset. The voicing is part of the encadeamento rule itself: the coverage level filters *which pairs* may fire, never the phonology of a firing rule.

Levels: **minimal** = encadeamento (voicing included) after the function-word class, editable in `data/pt_function_words.txt` (articles *o a os as um uma*, contractions *do da no na*, possessives, atonic pronouns, prepositions/conjunctions *de em por para com sem*, relatives and demonstratives); **standard** = after any word; **careful = standard**.

8.2 Worked derivations

"os amigos" → `o.Z@.mi.guS`.
os is a function word (minimal); `oS` splits into (`o`, `S`); the link key voices `S`→ `Z`; `o` + `.Z`+`@.mi.guS`. Same output at `minimal` and `standard`.

"mil amigos" → `mi.la.mi.guS` (standard only).
mil is lexical: at `minimal` the pair does not fire (`mil.a.mi.guS`); at `standard` the `l` links.

"as aves" → `@.Za.veS`.
Voicing `S`→`Z` (demo page).

“dos elefantes” → `du.Ze.le.f@~teS`.
Contraction + voicing.

“ser útil” → `se.Ru.til`.
`seR` → (`se`, `R`); the `R` migrates without voicing (only `/s/` voices). `pre` fails: the respelling would need the written accent (*rútil*) which the accent-less pt rule table cannot read — chain unlinked, documented.

“mar azul” → `ma.R@.zul`; “paz eterna” → `pa.Ze.te.Rna`.
Plain `R` migration; `-z` pausal `S`? no: *paz* → `pas`, the `s` voices to `z` and links.

“um elefante” — no-op.
Vocalised nasal `9~`: no movable consonant; elision and progressive nasalisation are out of scope (they would remove or nasalise a nucleus).

Expressive composition.
Os amigos e as aves with `--expressive`: two voiced encadeamentos, F0 tiers aligned.

Chapter 9

German — Glottalversatz, the Anti-Liaison

"ein Apfel" → `ajn a.pf@l`

The framework's deliberate mirror image (`sandhi/de.py`, after Kohler 1994 and Duden): where French inserts a consonant and Spanish displaces one, careful German *marks the word boundary* with a glottal stop (re-articulation) before a vowel-initial lexical word.

9.1 The rule and its engine encoding

The base g2p links *all* adjacent words with `.` (`ajn.a.pf@l` = [aɪ.na.pf@l] — smooth, non-standard). The module inserts the boundary only before **lexical** words with vowel onset (noun, verb, adjective, adverb). Function words with vowel initials (*ein*-forms, *ich, er, es, ihn, ihm, ihnen, ihr…, und, oder, aber, ob, in, im, an, am, auf, aus, über, unter, ohne, auch* — editable list `data/de_glottal_exempt.txt`) keep the smooth linkage of casual speech.

The glottal stop is *not* a phoneme of the engine inventory (`ALL_PHONEME_KEYS` has no glottal symbol), and the engine is never modified: the boundary is encoded as the engine's native *word boundary* — the syllable link `.` is replaced by a *space*, which the engine realises as a short re-articulating gap. That is the standard realisation of glottal boundaries in systems without a /ʔ phoneme.

9.2 Levels

- **minimal**: the categorical core — the boundary before the *noun that follows an article/determiner* (*ein Apfel*; determiner list: *der die das den dem des*, the *ein-* and *kein-*forms, possessives, *dieser*, *jeder*, *welcher*, *alle…*).
- **standard**: boundary before *every* vowel-initial lexical word.
- **careful**: hyperarticulation — the function-word exemption is *lifted*; the boundary appears before function words too.

9.3 Worked derivations

"ein Apfel" → `ajn a.pf@l`.
Pair (*ein*, *Apfel*): *Apfel* vowel-initial lexical word, not exempt; *ein* is a determiner → fires already at `minimal`; the `.` before `a.pf@l` becomes a space. Same output at `minimal` and `standard` (test `test_de_glottal_levels`).

“der alte Mann” → `deR a.lt@.man`.
Lexical adjective *alte* after the determiner.

“ich esse ein Apfel” → `iC E.s@.ajn a.pf@l`.
Two pairs, opposite outcomes: *esse* is a verb (lexical, vowel-initial) → boundary; *ein* after it is a function word → smooth linkage kept.

“das ist ein Ei”.
At `standard` the sentence carries the boundary before the noun *Ei*; at `careful` a second boundary appears before the function word *ein* — the test literally counts word-boundary spaces: one at `standard`, two at `careful` (test `test_de_careful_leve_exemption`).

“ein Adler” → `ajn a.dl@R`.
Lexical noun, vowel-initial.

“wie ein Baum” — unchanged.
Function word *ein* (exempt) plus a consonant-initial noun: no boundary at all — smooth linkage is the norm.

Expressive composition.
Ich esse ein Apfel und ein Ei with `--expressive`: glottal boundaries are laid down *after* the breathing plan — never inserted across a `%` pause.

`pre` is structurally inactive for German — *no German spelling encodes a glottal stop* — and says so with one warning per candidate boundary (documented no-op; use `g2p/post`).

Chapter 10

Coverage Levels: off / minimal / standard / careful

The level (option `--sandhi`, API parameter `level`) is a **monotone coverage filter** over each language's rule set: `minimal` ⊆ `standard` ⊆ `careful` (`sandhi/base.py`, rank table). It decides *which pair classes are allowed to fire*; it **never changes the phonology** of a rule that fires — e.g. Portuguese linking voicing applies identically at every level that lets the pair fire.

level	fr	en / it / es / pt	de
`minimal`	obligatory liaisons: determiners, pronouns, copulas/auxiliaries	en: *a*→*an*. it: euphonic *d* + monosyllabic RS triggers. es: y→e, o→u. pt: encadeamento (voicing incl.) after function words.	Glottalversatz after article/determiner
`standard` (API default)	+ everyday class: adverbs, prepositions, negation, preposed adjectives, modals, numerals	en: + onset linking / linking-r. it: + full RS (oxytons, come/dove/qualche). es: + enlace. pt: + everywhere.	+ before every vowel-initial lexical word
`careful`	+ style soutenu: liaison after plural noun/adjective (documented `-s`/`-x` heuristic, verbal endings excluded)	= standard (en, it, es, pt)	+ before function words too (hyper-articulation)

Table 10.1: Coverage matrix (from `README.md` / module docstrings).

Defaults and compatibility:

- The CLI defaults to `--sandhi off` (bit-identical baselines); the Python API defaults to `level='standard'`.
- The historical `liaison_*` functions (v1–v3) called *without* a level reproduce *exactly* the behaviour of the five-language patch (test `test_fr_v3_none_est_standard`); a `None` level is `standard`.
- The "= standard" cells are *assumed and documented*: where the language has no reliable

register-dependent rule at the token level, `careful` equals `standard` — never a silent difference.

- An unknown level raises `ValueError` immediately (a typo never silently degrades to `off`).

Chapter 11

Editable Tables (`data/*.txt`)

Each data file is an **editable view** over one embedded table of a module (`sandhi/base.py`, `load_wordlist`): one entry per line, lowercase, `#` comments, blank lines allowed.

file	table
`data/fr_h_aspire.txt`	*h aspiré* words blocking liaison (fr)
`data/it_rs_triggers.txt`	function-word triggers of raddoppiamento (it)
`data/pt_function_words.txt`	function words: minimal encadeamento (pt)
`data/de_glottal_exempt.txt`	function words exempt from the glottal boundary (de)

Loader semantics — priority *view over embedded table*:

- the file exists *and is non-empty* → its content **replaces** the embedded table;
- missing or unreadable file, or empty content → the embedded default applies; the package therefore always stays importable and testable — the txt files are **not** a dependency;
- a word *added* to `it_rs_triggers.txt` fires at level `standard`; the `minimal` subclasses are embedded in the modules (see each docstring);
- the level of an *existing* rule never changes: the level is a coverage filter, not a phonology dial (test `test_load_wordlist_fallback`: the shipped `fr_h_aspire.txt` replaces the fallback; an absent file falls back).

Chapter 12

Known Limitations (Documented, Never Silent)

Every item below is documented in the module docstrings and the CHANGELOG; each `pre`-mode gap emits a warning at runtime.

1. **`pre` mode — partial coverage by design.** It can only do what spelling encodes. French: measured respelling table, but no correct fallback for *est, petit, en, plus, vieux, ces, plein, jamais, beaucoup* (and the latent lands in coda, with small vowel shifts). Italian: E/e, O/o and Tuscan s/z rule-vs-lexicon tolerances; affricates, initial *q* and lexicon hiatus not encodable. Spanish: alternation is total, enlace respellings are verified but the palatal `R` has no spelling. Portuguese: atone vowel reduction tolerances (a/@, e/@, o/u); written accents break the round-trip. German and English: *no spelling at all* for the glottal boundary and the onset linking — structural no-op with warning.

2. **`post` mode — insertion cannot repair.** For French it inherits the base g2p per-word tokens (*il* → `i`: `fo.ti`). For it/es/pt/de/en the oracle is the identity and `post` equals `g2p`.

3. **Vowel-fusion phenomena are out of scope**: Spanish *sinalefa*, English *smoothing*, Portuguese atone elision and progressive nasalisation — all would delete or alter a syllabic nucleus and desynchronise the F0 plans of the expressive-prosody patch (the invariant of Section 3.5).

4. **French `careful` plural heuristic.** The noun/verb boundary is not marked in writing; the module uses the closed `-s`/`-x` heuristic, excluding verbal endings `-es -ent -ez -ons`.

5. **English module conventions.** It follows the pipeline's CMUdict gateway: British non-pre-vocalic `r` already present in the token (*far* → `faR`); flat tokens (no dynamic stress displacement).

6. **German boundary duration.** The glottal boundary is encoded as the engine's word-boundary space: its duration is the engine's inter-word gap, not an independent phoneme.

Chapter 13

Installation and Verification

13.1 The official procedure

The distribution ships the procedure as two equivalent scripts, `install.sh` and `install.ps1` (idempotent: an already-applied patch is detected and skipped; `-PipelineRef` can pin the verified baseline `41f6571`). The steps they chain:

1. **Clone the pipeline** (Branch A target):
   `git clone https://github.com/FBerthommier/covtl-pipeline.git`
2. **Apply the hook patch**: `git apply patch/covtl-pipeline-sandhi-hook.patch`
3. **Install the pipeline** as for the bare repository: `pip install -e .`
4. **Activate Branch B** — one of:
   `cp -r sandhi/. vtl_synth/utils/sandhi/` (copy, recommended; the trailing `/.` copies the *content* — nesting the folder one level too deep would leave the stub in place),
   or `pip install .` (site-packages resolution).
5. **Verify** (next section).

13.2 Verification

- **Package alone** (no pipeline needed):

  ```
  pip install pytest
  python -m pytest tests/ -q        # -> 38 passed
  ```

  The suite embeds g2p stubs carrying tokens *measured* on the pipeline g2ps; it checks the monotone levels per language, the three modes, the invariants (off = identity, no modification without a pair, never across a pause), the package API, the data files, and `decide_pair` on all six languages.

- **Hook alone** (clone @ `41f6571`): repository suite 155 → 165 passed (+10 hook tests), 3 pre-existing failures unchanged, **0 regressions**, outputs bit-identical.

- **Package copied**: repository suite unchanged plus 30/30 CLI end-to-end cases (6 languages × levels × modes, SAMPA verified); **pip** resolution verified (`fo.til`).

- **Smoke tests** after activation:

```
vtl-synth run "faut il"    -l fr --sandhi standard --no-video -o out  # fo.til
vtl-synth run "far away"   -l en --sandhi standard --no-video -o out  # fa.R@.we
vtl-synth run "los amigos" -l es --sandhi standard --no-video -o out  # lo.sa.mi.gos
vtl-synth run "os amigos"  -l pt --sandhi standard --no-video -o out  # o.Z@.mi.guS
vtl-synth run "a casa"     -l it --sandhi standard --no-video -o out  # a.kka.sa
vtl-synth run "ein Apfel"  -l de --sandhi standard --no-video -o out  # ajn a.pf@l
```

13.3 The Python API in one page

```
from sandhi import (SANDHI_LEVELS, sandhi_supported, apply_sandhi,
                    apply_sandhi_text, sandhi_rewrite_text,
                    apply_sandhi_sampa, sandhi_oracle, join_words_keys)

sandhi_supported('fr')                      # True (fr/en/it/es/pt/de)

# gateway over a text (monotone path - the hook entry point)
sampa = apply_sandhi_text('fr', 'faut il', g2p, level='standard',
                          mode='g2p')       # 'fo.til'

# low level: per-word keys, AFTER the per-word g2p (fusion in the slots)
keys = apply_sandhi('es', ['los', 'amigos'], ['los', 'a.mi.gos'],
                    'standard')             # ['lo', 'sa.mi.gos']
join_words_keys(keys)                       # 'lo.sa.mi.gos'

# expressive path: merge into an already planned, breath-marked SAMPA
sampa2 = apply_sandhi_sampa('fr', 'mon ami', planned_sampa, g2p,
                            level='standard', mode='g2p')

# pre mode alone: the rewritten spelling (to feed the planner)
sandhi_rewrite_text('en', 'a apple', g2p, 'minimal')   # 'an apple'
```

Each module additionally exposes the historical v3 API (`plan_pairs`, `liaison_oracle`, `apply_liaison_pre`, `apply_liaison_post`, `liaison_g2p`) and the covtl-sandhi aliases (`decide_pair`, `sandhi_g2p`, `apply_sandhi_pre`, `apply_sandhi_post`, `sandhi_oracle`).

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[15] Internal sources: `README.md` / `README.fr.md`, `INSTALL.txt`, `CHANGELOG.md` of `covtl-sandhi` v0.1.0; docstrings and tables of `sandhi/{fr,en,it,es,pt,de}.py` and `sandhi/base.py`; `tests/test_sandhi.py`; `README_LIAISON.md` of the five-language pack v3.